Towards Machine-Actionable Information Collection for Human Data Research and Beyond

Data Stewardship Wizard (DSW)

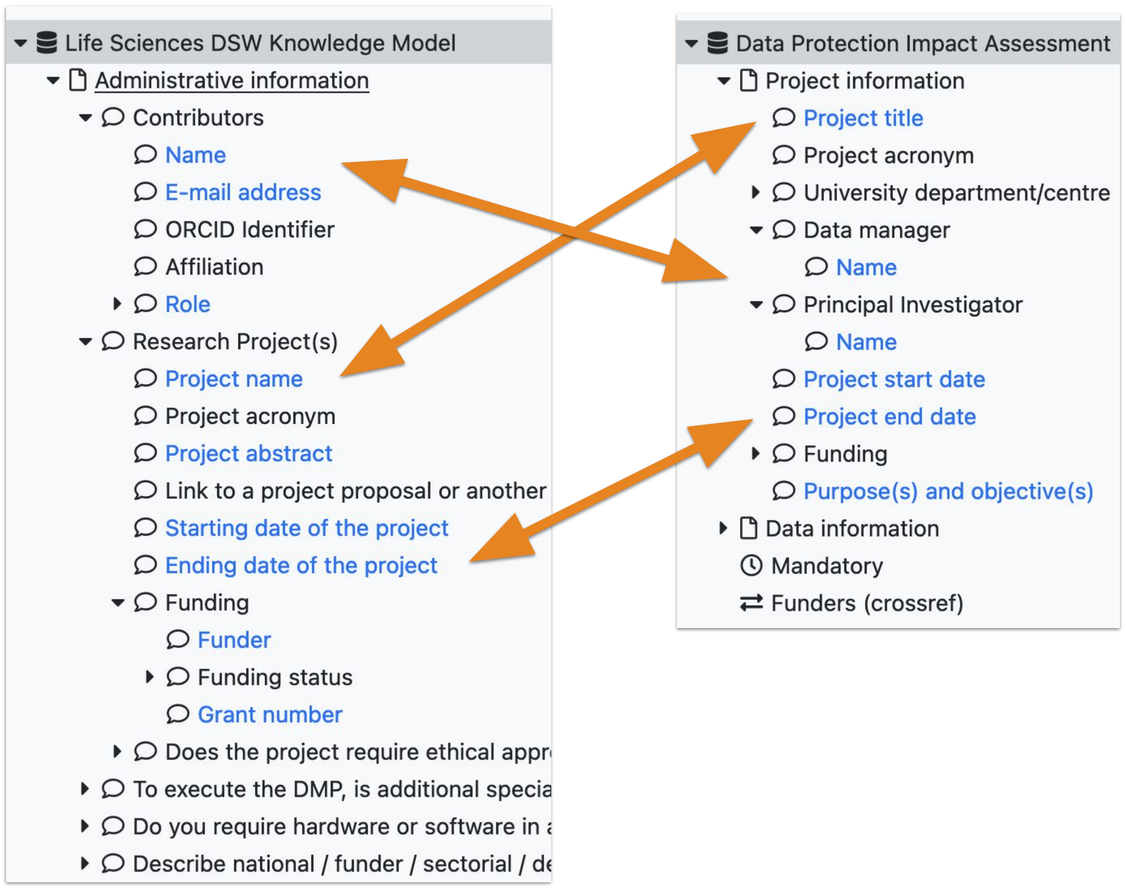
Research involving human data requires the creation of multiple documents with overlapping information (DMP, DPIA, ethics approval, consent form template, etc.). Additionally, researchers encounter difficulties in addressing GDPR-related inquiries. Data Stewardship Wizard (DSW) proves to be a valuable tool for researchers in executing such tasks, as it as it allows questions with the desired granularity, pre-defined answers and accompanying guideline. However, maintenance of several documents with overlapping information requires substantial efforts from data stewards.

Fact: multiple documents with repetitive information

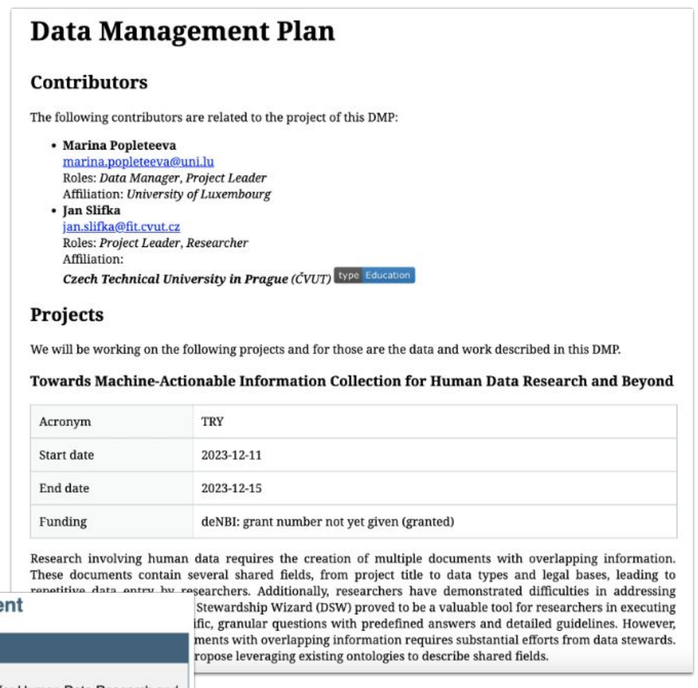
Issue: administrative burden for researchers and data stewards

Solution: transfer of identical information between documents in a few clicks

Annotated knowledge models in DSW



Document is filled by a researcher



Data Management Plan

Contributors

The following contributors are related to the project of this DMP:

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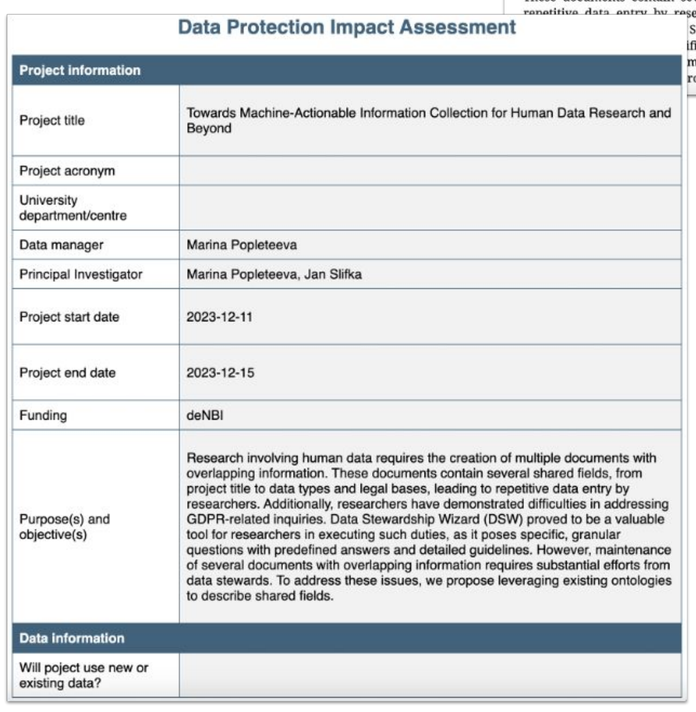
Projects

We will be working on the following projects and for those are the data and work described in this DMP.

Towards Machine-Actionable Information Collection for Human Data Research and Beyond

Acronym	TRY
Start date	2023-12-11
End date	2023-12-15
Funding	deNBI: grant number not yet given (granted)

Research involving human data requires the creation of multiple documents with overlapping information. These documents contain several shared fields, from project title to data types and legal bases, leading to repetitive data entry by researchers. Additionally, researchers have demonstrated difficulties in addressing GDPR-related inquiries. Data Stewardship Wizard (DSW) proved to be a valuable tool for researchers in executing specific, granular questions with predefined answers and detailed guidelines. However, maintenance of several documents with overlapping information requires substantial efforts from data stewards. To address these issues, we propose leveraging existing ontologies to describe shared fields.



Data Protection Impact Assessment

Project information	
Project title	Towards Machine-Actionable Information Collection for Human Data Research and Beyond
Project acronym	
University department/centre	
Data manager	Marina Popleteeva
Principal Investigator	Marina Popleteeva, Jan Slifka
Project start date	2023-12-11
Project end date	2023-12-15
Funding	deNBI
Purpose(s) and objective(s)	Research involving human data requires the creation of multiple documents with overlapping information. These documents contain several shared fields, from project title to data types and legal bases, leading to repetitive data entry by researchers. Additionally, researchers have demonstrated difficulties in addressing GDPR-related inquiries. Data Stewardship Wizard (DSW) proved to be a valuable tool for researchers in executing such duties, as it poses specific, granular questions with predefined answers and detailed guidelines. However, maintenance of several documents with overlapping information requires substantial efforts from data stewards. To address these issues, we propose leveraging existing ontologies to describe shared fields.
Data information	
Will poject use new or existing data?	

Shared fields are filled automatically

Process:

- Data stewards create knowledge models (KM) for each document.
- Shared fields are annotated in each KM with ontological key-value pairs.
- Researcher fills out one of the forms.
- Annotated fields in filled form are exported into RDF file using RDF document template.
- RDF project importer imports machine-readable data into the second DSW form with correct mapping.
- Researcher fills out the rest of the second form.

Resources:

- Annotated Life Science DSW and Data Protection Impact Assessment for Research knowledge models are available in registry.ds-wizard.org.
- RDF project importer is available at github.com/ds-wizard/dsw-rdf-importer.

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